

AI-Based Storage Tier Recommendation for Medical Image Repositories using Deep Learning Analytics

Research Gap Analysis and Literature Survey (Papers 11 to 15)

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1. What This Project Is About

Hospitals store a very large number of scans such as CT, MRI and X-ray. Keeping all of them on fast storage is expensive. Moving all of them to cheap storage is also a problem, because cheap storage can take hours to give a file back, and a doctor cannot wait that long.

Amazon S3 offers different storage classes. S3 Standard is fast but costly. Glacier Deep Archive is very cheap but slow. The aim of this project is to use deep learning to decide which storage class each scan should go into.

For this part of the literature survey I studied papers 11 to 15. For each paper I have written what the authors did, what data they used, what is good about it, what is missing, and what gap is left that our project can fill.

2. The Five Papers I Studied

#	Title	Journal	Year
P11	Machine learning-driven automatic storage space recommendation for object-based cloud storage system	Complex & Intelligent Systems (Springer)	2022
P12	Secure and fault tolerant cloud based framework for medical image storage and retrieval	Scientific Reports (Nature)	2025
P13	PARADIM: A Platform to Support Research at the Interface of Data Science and Medical Imaging	J. Imaging Informatics in Medicine (Springer)	2025
P14	A Guideline for Open-Source Tools to Make Medical Imaging Data Ready for AI Applications	J. Imaging Informatics in Medicine (Springer)	2024
P15	BVFLEMR: Federated Learning and Blockchain for a Cloud-Based Medical Records Recommendation System	Journal of Cloud Computing (Springer)	2022

3. Research Gap Analysis (Paper by Paper)

3.1 Paper 11 — Machine learning-driven automatic storage space recommendation for object-based cloud storage system

A. S. Mondal, A. Mukhopadhyay and S. Chattopadhyay, *Complex & Intelligent Systems*, vol. 8, pp. 489-505, 2022.

In one line: This paper uses machine learning to pick a storage location automatically, but it decides from server readings only and never looks at the file itself.

What the authors did: When data arrives at their cloud storage system, called RSoS, a monitoring tool named Ganglia watches the server while the data is being written and records 18 readings such as CPU, memory, input/output and load. A feature selection step keeps only the useful readings, and a classifier then predicts whether the incoming data is a sensor reading, a document or a file. Based on that prediction, the data is sent to the matching database: Cassandra, MongoDB or HDFS. They tried three feature selection methods (Fisher score, F-score and LII21) with three classifiers (SVM, K-NN and Neural Network), so nine combinations in total. LII21 with K-NN worked best.

Data they used: No public dataset was used. They made their own by running the storage system on one workstation and recording what the server did. Each row has the 18 server readings plus a label saying which of the three data types it was. The three types were kept in an equal 1:1:1 ratio. Health data was used as the example workload, including sensor streams such as pulse, temperature, SpO2, airflow and ECG, along with documents and files.

Advantages:

- It proves the main idea our project depends on: a machine learning model can decide by itself where a piece of data should be stored.
- Before this, a person had to edit a schema file by hand every time a new data type arrived. That manual step is removed completely.
- Sending each data type to the database that suits it reduces query time and helps save energy in the data centre.
- The features are server readings, not the data itself, so patient data is never opened. That is good for privacy.
- The nine-way comparison is useful for us. It tells us which feature selector and classifier to start with instead of guessing.

Limitations:

- Training is done offline. The authors say themselves that if the model learns something wrong, there is no way to correct it while the system is running.
- It knows only three classes: sensor, document and file. Medical images in DICOM format are not handled at all.
- Everything ran on one local workstation. It was never tested on a real cloud such as AWS, so we do not know how it behaves under real load.
- The dataset is small and made by the authors themselves, so nobody else can repeat the experiment or compare against it.
- It chooses a database, not a storage class. Cost is never calculated anywhere in the paper.
- Even though healthcare data is used, there is no encryption and no access control.
- It was published in 2022, which is outside the 2023 to 2026 range preferred in the guidelines. I have included it because it is the only paper that treats storage placement itself as a learning problem.

RESEARCH GAP

The model here decides where data goes by watching the server, not by looking at the data. For text and sensor readings that may be acceptable, but for medical images it is not. Two CT scans written to the same server at the same moment will produce almost the same CPU, memory and load readings, so this model

would treat them identically, even if one is a routine normal scan and the other is one a doctor will open again next week. The decision is also taken once, offline, and can never be corrected afterwards. And the choice is between databases, never between storage classes with different prices, so the question our project asks, which S3 class should this scan go into, is never asked here.

How we can improve on this: We keep the core idea and change what the model looks at. Instead of server readings, our model will look at the scan itself using deep learning, together with DICOM details such as scan type and body part, and the hospital's record of how often similar scans were opened again. The output will be an Amazon S3 storage class such as Standard, Standard-IA, Glacier Instant Retrieval or Deep Archive, instead of a database name, so the decision is tied directly to cost. We will also fix the offline problem the authors admit to, by checking the decision again through S3 Lifecycle rules so a scan can be moved later if it starts being opened more often.

3.2 Paper 12 — Secure and fault tolerant cloud based framework for medical image storage and retrieval in a distributed environment

A. Amaithi Rajan, V. Vetriselvi, M. Ajitesh and R. Praveen Kumar, Scientific Reports, vol. 15, art. 32965, 2025.

In one line: This paper protects and retrieves medical images across several clouds, but it keeps every scan in the same place at the same cost.

What the authors did: They build a framework called SFMedIR with four parts. First, federated learning: several hospitals train a shared model, based on a ConvNeXt network, that turns each image into a short code, without ever sending their images to one another. Second, the images are encrypted using a quantum-chaos method that is meant to stay safe even against future quantum computers. Third, a shadow scheme splits the stored data across one master cloud and several slave clouds, so nothing is lost if one cloud fails. Fourth, the whole system runs inside Docker containers and was deployed on Amazon EC2.

Data they used: Three public medical image datasets were used: Alzheimer Brain MRI with 5,120 training, 1,280 testing and 1,000 retrieval images; Brain Tumour MRI with 7,023 images across four classes; and Kidney CT with 10,000 training, 2,466 testing and 1,000 retrieval images. Attacks were simulated using the standard PGD method. Retrieval accuracy was about 90 percent overall, 5 to 10 percent better than earlier models, 60 to 70 percent better on attacked queries, and still above 75 percent while under attack.

Advantages:

- It shows that a deep learning model can turn a medical scan into a short code that describes what is inside it. Our project needs exactly this kind of image understanding.
- Federated learning means the hospital never has to send raw patient images anywhere, so our project can train the same way without breaking privacy rules.
- It handles security, fault tolerance and retrieval accuracy together, where most papers handle only one of the three.
- It was actually deployed on Amazon EC2, so the design is not purely theoretical.
- There is a formal security proof as well as experiments, and three separate datasets were used, so the results are not tied to one collection of images.

Limitations:

- The experiments assume a perfect world: honest servers, secure channels, and a stable network with almost no packet loss. A real hospital network is not like this.
- It was only a small Docker simulation on EC2, not a real deployment across regions over a long period.
- Encryption, federated training and the shadow scheme are all heavy work, but the paper never says what this costs in money or in extra waiting time for the user.
- Every image is treated the same way. There is no idea of a frequently used scan and a rarely used one, so there is no cheap storage anywhere in the design.
- Copying data across master and slave clouds actually multiplies the storage bill, and that bill is never reported.
- It builds its own security instead of using ready-made cloud services such as AWS KMS and IAM, which makes it harder to certify against hospital rules.
- No radiologist checked whether the images it retrieves are actually good enough to make a diagnosis.

RESEARCH GAP

This paper makes storage safe and reliable but never makes it cheap. Every scan is encrypted, split and copied across clouds in exactly the same way, whether it is opened every day or never again. The short code the model produces describes what the image contains, and that is precisely the information needed to guess whether a scan will be wanted again. But it is used only for searching. It is never sent back to the storage side to decide where the scan should live. So the paper builds the exact tool our project needs and then uses it for a different purpose.

How we can improve on this: We will take the image code idea and point it at a different question. The same code this paper uses for searching can also be used to predict how likely a scan is to be opened again, and therefore which S3 class it belongs in. Cheap protection can come from AWS itself, using S3 with KMS encryption and Cross-Region Replication instead of a custom shadow scheme, and only the important scans need the expensive replication. Most importantly, we will report the monthly bill and the time taken to get a scan back, which this paper leaves out completely.

3.3 Paper 13 — PARADIM: A Platform to Support Research at the Interface of Data Science and Medical Imaging

Y. Lemarechal et al., Journal of Imaging Informatics in Medicine, vol. 39, pp. 1783-1795, 2025.

In one line: This is a complete working platform for hospital scans, but all the scans sit on the same storage and nothing ever decides to move them.

What the authors did: They built a platform called PARADIM out of free open-source parts, and it is in real use today. Karnak removes patient identity from the scans while they are still inside the hospital. Kheops organises the scans into collections and controls who can see them. Orthanc is the actual store where the DICOM images are kept. Keycloak handles logins with multi-factor authentication. The OHIF viewer lets researchers look at and label images. An in-house Job-Manager running on Kubernetes launches heavy jobs such as automatic segmentation, dose calculation and AI

model testing. Everything stays in DICOM format from start to finish, and every step is recorded so results can be traced and repeated.

Data they used: This is not an experiment with a dataset. The platform holds real medical image collections belonging to the institution, including CT, MRI, radiotherapy plans and dose maps, and is used by around ten research projects that have ethics approval.

Advantages:

- It shows a complete working pipeline: bring the scan in, remove patient identity, store it, label it, train on it and publish the result. Our project has to fit inside a pipeline like this one.
- Keeping everything in DICOM means details such as body part, scan type and acquisition settings are never lost. Our model needs exactly those details as input features.
- Security and governance are handled properly, with logins, multi-factor authentication, permissions and a full record of who did what.
- It is built from mature free components, so licensing cost is low and any part can be replaced.
- It directly attacks the reproducibility problem caused by datasets of unknown origin, which is a real weakness in medical AI.

Limitations:

- It runs on the institution's own machines. When a big job arrives there is no way to borrow extra capacity, and when the machines are idle the money is still spent.
- The platform moves data around but never thinks about it. There is no model that decides where a scan should be kept or when it can be moved to cheaper storage.
- All scans sit on the same storage. There are no tiers, no archive and no cost figures anywhere in the paper.
- The authors note that outside analysis tools still need the data to be downloaded locally, which weakens the security boundary they worked hard to build.
- It handles DICOM images only. Patient records, lab results and doctors' notes cannot be stored in it.
- Running it needs real engineering skill in Kubernetes, OpenShift and Orthanc plugins, which a small hospital or a student team will not have.
- Labelling is still partly manual. Web-based contouring is described as work that is still going on.

RESEARCH GAP

PARADIM is the best example of how hospital images should be organised, and it is exactly the kind of place where our project would sit. But it has no cost thinking at all. Every scan, whether it was viewed fifty times this month or last opened four years ago, is kept on the same storage at the same price. The platform already records everything needed to make a better decision, namely who opened which scan, when, and how often. That record is used only for tracing the research. It is never fed back to decide where the scan should live, and there is no automatic archiving, so old scans stay on expensive storage for ever.

How we can improve on this: We will follow the same structure but place it on AWS so storage can be tiered and cost can be controlled: Amazon S3 for the images, AWS Lambda for de-identification, Amazon Cognito in place of Keycloak, and Amazon SageMaker for training. Then we add the missing piece, which is a deep learning model that reads the scan and the access history PARADIM already

collects, and predicts which S3 class the scan belongs in. S3 Lifecycle rules then carry out the move automatically, so no person has to do it.

3.4 Paper 14 — A Guideline for Open-Source Tools to Make Medical Imaging Data Ready for AI Applications: A SIIM Survey

S. Vahdati et al., Journal of Imaging Informatics in Medicine, vol. 37, pp. 2015-2024, 2024.

In one line: This paper lists the free tools that get medical images ready for AI, but it never asks where those images should be stored or what that costs.

What the authors did: This is a survey, not a system. A questionnaire was sent to 500 members of the Society of Imaging Informatics in Medicine, asking which free tool they use for preparing medical images and why. 54 replies came back from 26 medical informatics centres. After removing repeats, vague answers and tools that were not publicly available, 28 tools were left. Each tool was then described by what it can do, including removing patient identity, viewing images, working in the cloud, converting formats, normalising, labelling, annotating, segmenting, storing and supporting federated learning, and the tools were grouped by which stage of preparation they belong to.

Data they used: The survey itself is the data: 54 responses from 26 centres, reduced to 28 tools such as 3D Slicer, the RSNA Anonymizer, CTP, DIANA, dcm2niix, ITK-SNAP, MITK, MONAI Label, MIDRC, Niffler and CVAT. No medical images were analysed at any point.

Advantages:

- It is genuinely useful for us. It saves weeks of searching when we have to choose a tool to remove patient identity or to convert DICOM files before feeding them to our model.
- It covers the whole preparation pipeline rather than one step, so we can see how the stages connect to each other.
- The answers come from people working in 26 real informatics centres, not from advertising material.
- It marks which tools work in the cloud and which support federated learning, which matters directly for a cloud project like ours.
- It warns about specific traps, such as tools that can only produce one label when several labels are needed.

Limitations:

- Nothing is measured. There are no numbers for speed, memory or accuracy, so we still cannot say which tool is best, only which ones are popular.
- Only 54 of 500 people replied, about 11 percent, and all from one society, so the list may not reflect world practice.
- Tools change quickly, so the list goes out of date. The authors say this themselves.
- Paid tools and in-house tools were left out, even though many hospitals depend on them.
- The tools are listed one by one. There is no design showing how to join them into an automatic pipeline.
- Cost, security and legal compliance of running these tools at scale are not examined.

- Working in the cloud is recorded as a simple yes or no. Which cloud, and at what cost, is never discussed.

RESEARCH GAP

This paper stops at the exact moment our project starts. It explains carefully how to get an image ready for a model, and then says nothing about where the image goes afterwards. Storage appears only as a tick-box feature of a tool, never as a decision with a price attached. None of the 28 tools looks at how often a scan is opened, none of them chooses a storage class, and none of them can move a file to cheaper storage later. Worse, the information our model needs is actually created during preparation, such as scan type, body part and image quality, but it is thrown away once the image has been converted.

How we can improve on this: We will use the tools this survey recommends, but chain them into an automatic pipeline instead of running them by hand: a scan lands in Amazon S3, an event starts AWS Step Functions, and AWS Lambda runs de-identification and format conversion. At the same time we will keep the details these tools produce, such as scan type, body part, file size and image quality, and pass them into our tier model as features instead of discarding them. We will also record the time taken and the cost per scan, which is the number this survey never gives.

3.5 Paper 15 — BVFLEMR: An Integrated Federated Learning and Blockchain Technology for Cloud-Based Medical Records Recommendation System

T. Hai, J. Zhou, S. R. Srividhya, S. K. Jain, P. Young and S. Agrawal, Journal of Cloud Computing, vol. 11, art. 22, 2022.

In one line: This paper decides what to do with patient records using machine learning, but it works on text records and keeps everything stored the same expensive way.

What the authors did: The system has two parts. In the first, electronic health records are written into a Hyperledger Fabric blockchain that sits on a cloud server, so every change to a record is permanently recorded and cannot be secretly altered. In the second, vertical federated learning collects information from several of these blockchains, and a LightGBM model together with an N-Gram language model and sentiment analysis recommends a suitable treatment for the patient.

Data they used: An electronic health record dataset was used together with a free sentiment and emotion library. The paper never says which dataset it is, how large it is, or where it came from. The data was fed in step by step at 20, 40, 60, 80 and 100 percent. Average precision was 0.953, recall 0.932 and F1 score 0.950. LightGBM is said to be around 20 times faster than similar gradient boosting models.

Advantages:

- It shows that decisions about healthcare data can be treated as a machine learning problem, which is the same assumption our project makes.
- LightGBM is fast and light on memory, so it is realistic for a student project. Our tier model can use it as a baseline before trying anything heavier.
- Every change to a record is tracked, which gives a natural audit trail for hospital rules.
- Precision, recall and F1 score are all reported, not only accuracy, so the results can be judged properly.
- Federated learning is used, so hospitals never have to hand over their raw records.

Limitations:

- The dataset is not named, not sized and not sourced. Nobody can repeat the work or compare against it.
- The results get worse as more data is added. Recall falls from 0.952 at 20 percent to 0.920 at 100 percent, and the paper never explains why.
- Blockchain is expensive and slow, but the paper never measures write speed, delay or storage cost. These are the main reasons hospitals hesitate to use it.
- Only text and tabular records are handled. Medical images, which take up most of the storage in a hospital, are not covered at all.
- Everything is stored in the same way. There is no cheap storage, no archive and no idea that some records are needed less often than others.
- The treatment recommendations were never checked by a doctor, and the model cannot explain why it recommended anything.
- It was published in 2022, which is outside the 2023 to 2026 range preferred in the guidelines. I have included it because it is the closest work to the idea of learning where healthcare data should be kept.

RESEARCH GAP

This paper puts intelligence in front of storage, which is the right idea, but the storage behind it has only one setting. A blockchain keeps every record permanently and at full cost, so a record nobody has opened in five years costs exactly as much as one being used today. The decision is also based on the text of the record, and no image is ever examined. For an image archive, where a single CT study can be hundreds of megabytes, this design would be extremely expensive. Nothing here predicts whether a record will be needed again, and that is the one prediction that decides the bill.

How we can improve on this: We will keep the useful part, which is treating the storage decision as a learning problem and using LightGBM as a fast baseline, but we will change what it decides. Our model works on one scan at a time, using features taken from the image itself plus its access history, and it chooses an Amazon S3 storage class instead of storing everything permanently. Only a small hash of each scan needs to be kept in a tamper-proof ledger such as Amazon QLDB, while the scan itself sits in the S3 class the model chose. We will also report the cost and the time taken to get a scan back, which this paper never does.

4. Overall Research Gap

Putting all five papers together, this is what I found:

- Paper 11 shows that a model can choose a storage location on its own, but it decides from server readings and never looks inside the file, and it picks a database rather than a priced storage class.
- Paper 12 builds a deep learning model that describes what is inside a scan, but uses that description only for searching, and stores every scan the same expensive way.
- Paper 13 gives a complete, working platform for hospital images, but keeps every scan on one type of storage and never decides to move anything.
- Paper 14 explains how to get images ready for AI, but treats storage as a tick box and never mentions cost or storage class.

- Paper 15 does treat a healthcare data decision as a learning problem, but works on text records, keeps everything permanently, and never measures what that costs.

So the gap is the same one seen from five directions. Every one of these papers either understands the image but ignores the storage bill, or manages the storage but never looks at the image. No existing work reads the medical image itself with deep learning, combines that with the record of how often the scan was opened before, and uses the result to predict whether the scan will be needed again and place it in the correct Amazon S3 storage class. That is exactly what our project will do.

5. Literature Survey Table

Paper	Method	Dataset	Advantages	Limitations	Research Gap
P11 — Mondal et al. (2022), Complex & Intelligent Systems, Springer	Ganglia records 18 server readings (CPU, memory, I/O, load) for each write. Feature selection (Fisher, F-score, LII21) plus a classifier (SVM, K-NN, Neural Network) predicts if the data is sensor, document or file, then sends it to Cassandra, MongoDB or HDFS. LII21 with K-NN was best.	Own data from running RSoS on one workstation. 18 server readings per write plus a label, three classes in 1:1:1 ratio. Health workload of sensor streams (pulse, temperature, SpO2, airflow, ECG), documents and files.	Proves a model can choose a storage location by itself. Removes a manual schema edit. Matching data type to database cuts query time and saves energy. Patient data is never opened. Useful nine-way comparison of selectors and classifiers.	Training is offline and cannot be corrected at run time. Only three classes, no DICOM. Tested on one workstation, not a real cloud. Small self-made dataset. Picks a database, not a storage class. No cost calculation, encryption or access control.	Two CT scans written at the same moment give almost the same server readings, so an important scan and a routine one are treated alike. The decision is taken once, offline, and is between databases rather than priced storage classes.
P12 — Amaithi Rajan et al. (2025), Scientific Reports, Nature	SFMedIR: federated learning with a ConvNeXt network turns each image into a short code without sharing images; quantum-chaos encryption protects the images; a shadow scheme spreads data across master and slave clouds for fault tolerance. Run in Docker on Amazon EC2.	Alzheimer Brain MRI (5,120/1,280/1,000), Brain Tumour MRI (7,023 images), Kidney CT (10,000/2,466/1,000). PGD attacks simulated. About 90% retrieval accuracy, 5-10% above earlier models, 60-70% better on attacked queries.	Shows a model can turn a scan into a short code describing its content. Federated learning keeps raw images inside the hospital. Handles security, fault tolerance and accuracy together. Deployed on Amazon EC2. Formal security proof plus three datasets.	Assumes honest servers and a near-perfect network. Only a small Docker simulation on EC2. Cost and extra delay never measured. Every image treated the same, so no cheap storage. Copying across clouds multiplies the bill. Custom security instead of AWS KMS and IAM.	Storage is made safe and reliable but never cheap. The image code describes what is inside a scan, exactly what is needed to guess if it will be wanted again, but it is used only for searching and never sent back to the storage side.
P13 — Lemarechal et al. (2025), J. Imaging Informatics in Medicine, Springer	PARADIM platform built from open-source parts: Karnak removes patient identity, Kheops manages collections and access, Orthanc stores the DICOM images, Keycloak handles login with MFA, OHIF is used for labelling, and an in-house Job-Manager on Kubernetes runs heavy jobs.	Not an experiment. Holds real institutional collections of CT, MRI, radiotherapy plans and dose maps, used by about ten ethics-approved research projects.	A complete working pipeline from ingest to publication. Staying in DICOM keeps scan type, body part and settings, which our model needs as features. Proper login, MFA, permissions and full traceability. Built from free components.	Runs on the institution's own machines, so no extra capacity when needed and wasted money when idle. Moves data but never reasons about it. All scans on one storage: no tiers, no archive, no cost figures. DICOM only. Needs heavy DevOps skill.	Every scan costs the same whether it was viewed fifty times this month or last opened four years ago. The platform records who opened which scan and how often, but uses it only for tracing research, never to decide where the scan should live.
P14 — Vahdati et al. (2024), J. Imaging Informatics in Medicine, Springer	A survey, not a system. Questionnaire sent to 500 SIIM members, 54 replies from 26 centres, reduced to 28 open-source tools. Each tool described by what it can do (de-identify, view, cloud, convert, normalise, label, annotate, segment, store, federated learning) and grouped by preparation stage.	The survey itself: 54 responses from 26 medical informatics centres, reduced to 28 tools such as 3D Slicer, RSNA Anonymizer, CTP, DIANA, dcm2niix, ITK-SNAP, MITK, MONAI Label, MIDRC, Niffler and CVAT.	Saves weeks when choosing a de-identification or conversion tool. Covers the whole preparation pipeline. Answers come from 26 real informatics centres. Marks which tools work in the cloud and which support federated learning.	Nothing measured, so no evidence of which tool is best. Only about 11% replied, all from one society. List dates quickly. Paid and in-house tools excluded. Tools listed separately with no automatic pipeline. Cloud support is a yes/no flag.	It explains how to prepare an image and then says nothing about where it goes. No tool tracks how often a scan is opened, chooses a storage class, or moves a file later. Details created during preparation are thrown away instead of reused.
P15 — Hai et al. (2022), Journal of Cloud Computing, Springer	BVFLEMR: health records are written into a Hyperledger Fabric blockchain on a cloud server so every change is permanent and traceable. Vertical federated learning then collects from several blockchains and a LightGBM model with an N-Gram language model and sentiment analysis recommends treatment.	An unnamed EHR dataset plus a free sentiment and emotion library. Size, source and feature count are not reported. Fed in at 20/40/60/80/100%. Precision 0.953, recall 0.932, F1 0.950. LightGBM said to be about 20 times faster.	Treats a healthcare data decision as a machine learning problem, the same assumption our project makes. LightGBM is fast and light, usable as our baseline. Every record change is auditable. Reports precision, recall and F1, not just accuracy.	Dataset unidentified, so not reproducible. Results get worse as data grows (recall 0.952 to 0.920) with no explanation. Blockchain speed, delay and storage cost never measured. Text records only, no images. Everything stored the same way with no cheap tier.	Intelligence sits in front of storage that has only one setting. A record untouched for five years costs the same as one in daily use. The decision reads text, never an image, and nothing predicts whether the data will be needed again.